

Generation-0. The earliest night vision devices existed during World War II and the 1950s. These devices were based on image conversion rather than image intensification. The night vision devices primarily comprised a photocathode that converted incident photons into electrons. The electrons were accelerated towards an anode by applying a positive potential to the anode. The devices also had an IR source of radiation, called IR illuminator, mounted on the device. The IR illuminator irradiated the target scene with IR radiation. IR radiation reflecting off the target back to the night vision devices was collected by their objective lens and focused on to the photocathode.

Generation-1. Generation-1 night vision devices were an adaptation of generation-0 technology—both used a photocathode and an anode, the former for photon-to-electron conversion and the latter to accelerate photo electrons towards it. A major deviation in generation-1 devices from generation-0 ones was the absence of an IR source that was used in the case of the latter devices to provide scene illumination.

Generation-2. These vision devices were the first to use an MCP for electron multiplication. This led to significant increase in device sensitivity.

Consequent light amplification was of the order of 20,000x. This resulted in much improved performance even in very low-light level ambient conditions of cloudy and moonless nights.

Introduction of the MCP into the intensifier tube also obviated the need to connect the tubes in series, as was done in generation-1 devices. This significantly reduced the size of the night vision devices, and made handheld devices and helmet-mounted devices a reality.

Generation-3. Compared to generation-2, generation-3 night vision devices have two distinctive changes: use of gallium arsenide photocathode and ion barrier coating on the MCP. Due to its higher quantum efficiency or radiant sensitivity, and its spectral response extending to near IR region, gallium arsenide photocathode enables target detection at longer ranges and in darker conditions.

Light amplification factor increased from 20,000 times in generation-2

devices to a maximum of 50,000 times in generation-3 devices. The ion barrier film improved the tube life from 2000 hours in generation-2 devices to 10,000 in generation-3 devices—even though it was at the cost of slight reduction in radiant sensitivity due to fewer photo electrons being able to reach the MCP.

Generation-3+ devices offer improved performance specifications over generation-3 devices. Two important features associated with generation-3+ night vision devices are an automatic gated power supply system and a thinned ion barrier layer. Absence of the ion barrier layer or its thinning improves luminous sensitivity, even though it is at the cost of a slight reduction in the life of the tube.

Another common term encountered while discussing night vision technology is omnibus, or OMNI. This refers to the multi-year/multi-product contracts of the US Army for the procurement of night vision devices from Exelis (formerly ITT Night Vision). Under these contracts, the company delivers generation-3 devices with increasingly higher performance. Current contract is for OMNI-VIII.

Generation-4. As of now, there is nothing like generation-4 night vision technology. Generation-4 night vision devices were initially conceived to use filmless and gated technology. The proposal was to remove the ion barrier film from the MCP, which was introduced in generation-3 devices. Removal of film was aimed at reducing background noise and enhance signal-to-noise ratio. It would also allow more electrons to reach the MCP, so that images were significantly less distorted and brighter.

Introduction of automatic gated power supply for the photocathode enabled the devices to adapt instantaneously to light level fluctuations from low-light level to high-light level, or from high-light level to low-light level. Removal of the ion barrier was also intended to reduce the halo effect seen around bright spots or light sources. While device performance improved, absence of the ion barrier film led to increased tube failure rates. As a consequence of this, the idea of film removal was abandoned in favour of a thinned film, giving birth to genera-



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